Rishabh Patidar

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Education

Madhav Institute of Technology and Science Gwalior, MP
Bachelor of Technology in Information Technology | ***CGPA: 8.124*** *Expected: May 2027*
Relevant Coursework: Machine Learning, Artificial Intelligence, Data Structures & Algorithms, DBMS, Linear Algebra

Technical Skills

Languages: Python, Java, C/C++, SQL
ML / AI: Scikit-learn, XGBoost, Random Forest, Gradient Boosting, KMeans, PCA, LLaMA-3.3-70B, Prompt Engineering, Vector Embeddings, Feature Engineering, SHAP, SMOTE, Hyperparameter Tuning
Computer Vision: InsightFace (ArcFace 512-d embeddings), MediaPipe (pose/gait descriptors), OpenCV, Weighted Cosine Similarity, Person Re-identification
Libraries: NumPy, Pandas, Matplotlib, Seaborn, Plotly
Deployment & MLOps: FastAPI, Streamlit, Pydantic V2, Groq API, Git, Docker (basic), Jupyter

Experience

Undergraduate Research Assistant Jan – Apr 2026
Madhav Institute of Technology and Science *Gwalior, MP*

- Architected a two-stage XGBoost Regressor pipeline for event footfall prediction by engineering venue and marketing features with cross-validated hyperparameter tuning via GridSearchCV, achieving $R^2 = 0.96$ on held-out test data
- Designed a Dynamic Physical Clamping algorithm by introducing hard venue-capacity constraints at inference time, eliminating all physically invalid predictions and ensuring production-safe model outputs across 100% of test cases
- Integrated LLaMA-3.3-70B via Groq API with Zero-Trust key validation to perform semantic brand-event synergy scoring and generate structured JSON negotiation reports, deployed as a stateless FastAPI microservice for production use

Projects

Video Target Identification System | *Python, InsightFace, MediaPipe, OpenCV, Streamlit*

- Architected a dual-model biometric fusion pipeline by combining InsightFace 512-d ArcFace embeddings and MediaPipe 12-d pose/gait descriptors via tuned weighted cosine similarity (face 0.7, pose 0.3), enabling robust person re-identification across multi-camera CCTV footage
- Reduced false positive rate by engineering consecutive-frame temporal validation with configurable score thresholds, ensuring identifications are confirmed across multiple frames before being accepted as a match
- Deployed the end-to-end system as a Streamlit forensic dashboard supporting multi-video batch scanning, real-time similarity score visualisation, and automated evidence export (PDF report, CSV log, ZIP of face crops) for downstream investigative use

Customer Segmentation & Churn Prediction | *Scikit-learn, XGBoost, SHAP, SMOTE, Streamlit*

- Engineered 22 RFM and behavioural features from 2.5M+ transactions by building a modular feature pipeline with SMOTE-based oversampling to address class imbalance, resulting in a Random Forest classifier achieving ROC-AUC 0.8793
- Applied KMeans (k=4) clustering with elbow-method optimisation to identify 4 statistically distinct customer segments, including a high-risk cohort with a 15.7% churn rate, enabling targeted downstream ML model training per segment
- Leveraged SHAP TreeExplainer to produce feature attribution plots confirming recency as the dominant churn predictor, translating model probability outputs into a $233,917 revenue-at-risk estimate across 392 flagged households

CinemaIQ — Box Office Intelligence Platform | *Python, XGBoost, Scikit-learn, Streamlit, Groq API*

- Designed novel actor, director, and composer power-index features using weighted historical performance aggregation and multi-hot genre encoding, combined with log1p target transformation to normalise revenue distribution and improve model fit
- Conducted a rigorous 5-model benchmark (Linear Regression, Ridge, Random Forest, Gradient Boosting, XGBoost) with consistent train/val/test splits, selecting XGBoost on lowest RMSE and superior holdout generalisation for South Asian film revenue prediction
- Deployed a production-ready 3-tab Streamlit dashboard with Pessimistic/Base/Optimistic scenario simulation, ROI gauges, and LLaMA 3.3 AI commentary via Groq API, enabling end-to-end revenue forecasting from raw feature input to actionable insight

Achievements

1st Runner-Up — SSH '26 National Hackathon | SSH | Feb 2026
Top Performer in WebDev Track — Hacksagon | ABV IIITM Gwalior | Apr 2026
Finalist — ABV-IIITM Hackatron | ABV-IIITM (Powered by GitHub) | Oct 2025